

Concept Note #15

PHP Add and Select Data

Insert data into a Database from an HTML Form

Step 1: Creating the HTML Form

Here's a simple HTML form that has three text `<input>` fields and submit button.

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title></title>
</head>
<body>
  <form action="samplerc.php" method="post">
    <p>
      <label for="firstName">First Name:</label>
      <input type="text" name="first_name" id="firstName">
    </p>

    <p>
      <label for="lastname">Last Name:</label>
      <input type="text" name="last_name" id="lastName">
    </p>

    <p>
      <label for="emailAddress">Email Address:</label>
      <input type="text" name="email" id="emailAddress">
    </p>
    <input type="submit" name="btn" value="Add Records">
  </form>
</body>
</html>
```

First Name:

Last Name:

Email Address:

Step 2: Retrieving and inserting the Form Data.

When a user clicks the submit button of the add record HTML form, in the example above, the form data is sent to 'insert.php' file. The 'insert.php' file connects to the MySQL database server, retrieves forms fields using the `$_REQUEST` variables and finally execute the insert query to add the records. Here is the complete code of our 'insert.php' file:

```
<?php

$link = mysqli_connect("localhost", "root", "", "sample");

if ($link === false) {
    die("ERROR: Could not connect." . mysqli_connect_error());
}

$first_name = mysqli_escape_string($link, $_REQUEST['first_name']);
$last_name = mysqli_escape_string($link, $_REQUEST['last_name']);
$email = mysqli_escape_string($link, $_REQUEST['email']);

$sql = "INSERT INTO sample (first_name, last_name, email) VALUES ('$first_name', '$last_name', '$email')";

if(mysqli_query($link, $sql)) {
    echo "Records added susccessfully.";
}
else {
    echo "ERROR: Could not able to execute $sql" . mysqli_error($link);
}

mysqli_close($link);
?>
```

Selecting Data from Database Tables

So far you have learnt how to create database and table as well as inserting data. Now it's time to retrieve data what have inserted in the preceding tutorial. The SQL `SELECT` statement is used to select the records from database tables. It's basic syntax is as follows.

SELECT column1_name, column2_name, column3_name **FROM** table_name;

Let's make a SQL query using the **SELECT** statement, after that we will execute this SQL query through passing it to the PHP `mysqli_query()` function to retrieve the table data.

Consider our persons database table has the following records:

	id	first_name	last_name	email
☐ Edit Copy Delete	1	Miguel	Batong-bakal	ironmigz@gmail.com
☐ Edit Copy Delete	2	Juan	Dela Cruz	dckimsy@gmail.com
☐ Edit Copy Delete	3	Maria	Palma	mariapalm@gmail.com
☐ Edit Copy Delete	4	Miguel	Batong-bakal	ironmigz@gmail.com

The PHP code in the following example selects all the data stored in the sample table (using the asterisk (*) in place of column name selects all the data in the table).

```
<?php
$link = mysqli_connect("localhost", "root", "", "sample");

if ($link === false) {
    die("ERROR: Could not connect." . mysqli_connect_error());
}

$sql = "SELECT * FROM sample";

if($result = mysqli_query($link, $sql)) {
    if(mysqli_num_rows($result) > 0) {
        echo "<table>";
        echo "<tr>";
        echo "<th>id</th>";
        echo "<th>first_name</th>";
        echo "<th>last_name</th>";
        echo "<th>email</th>";
        echo "</tr>";

        while ($row = mysqli_fetch_array($result)) {
            echo "<tr>";
            echo "<td>" . $row['id'] . "</td>";
            echo "<td>" . $row['first_name'] . "</td>";
            echo "<td>" . $row['last_name'] . "</td>";
            echo "<td>" . $row['email'] . "</td>";
            echo "</tr>";
        }
        echo "</table>";
        mysqli_close($link);
    } else {
        echo "No records matching your query were found.";
    }
} else {
    echo "ERROR: Could not be able to execute $sql." . mysqli_error($link);
}

mysqli_close($link);
?>
```

Explanation of Code (Procedural style)

In the example above, the data return by the `mysqli_query()` function is stored in the `$result` variable.

Each time `mysqli_fetch_array()` is invoked, it returns the next row from the result set as an array.

The while loop is used to loops through all the rows in the result set. Finally the value of individual field can be accessed from the row either by passing the field index or field name to the `$row` variable like:

`$row['id']` or `$row[0]`, `$row['first_name']` or `$row[1]`, `$row['last_name']` or `$row[2]`, and `$row['email']` or `$row[3]`

If you want to use the for loop you can obtain the loop encounter value or the number of rows returned by the query by passing the `$result` variable to the `mysqli_num_rows()` function.

This loop counter value determines how many times the loop should run.